

Predicting LLM Knowledge Performance from Reasoning Benchmarks: A From-Scratch Logistic Regression Approach

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Abstract—Large language models (LLMs) are routinely evaluated on diverse benchmarks spanning reasoning, mathematics, and factual knowledge. However, it remains an open question whether strong performance on multi-step reasoning tasks generalises to graduate-level knowledge benchmarks. This paper presents a binary classifier that predicts whether an LLM achieves above-median performance on the Graduate-Level Google-Proof Q&A (GPQA) benchmark using only its scores on reasoning-oriented tasks as input features. A logistic regression model implemented entirely from scratch in NumPy is trained on data from 995 models drawn from the Open LLM Leaderboard v2. Through a systematic grid search over learning rate and L2 regularisation strength, the best model achieves 91.33% test accuracy, an F1 score of 0.9091, and an ROC-AUC of 0.9712, substantially outperforming a rule-based threshold baseline (50% accuracy, AUC 0.50). Feature weight analysis reveals that MATH-Hard accuracy is the strongest positive predictor of GPQA performance, while MMLU-Pro accuracy is a strong negative predictor, suggesting that models specialising in breadth of factual recall do not necessarily excel at deeper knowledge reasoning.

Index Terms—large language models, benchmark evaluation, logistic regression, binary classification, reasoning vs. memorization, Open LLM Leaderboard

I. INTRODUCTION AND BACKGROUND

A. Motivation

The rapid proliferation of LLMs has been accompanied by an equally rapid expansion of evaluation benchmarks. Leaderboards such as the Open LLM Leaderboard v2 [1] aggregate scores across tasks ranging from logical deduction (BIG-Bench Hard [2]) and mathematical reasoning (MATH [3]) to factual recall (MMLU-Pro [5]) and graduate-level scientific reasoning (GPQA [6]). Despite this wealth of evaluation data, the relationship between performance on reasoning benchmarks and performance on deep knowledge benchmarks is not well understood.

A recurring concern in the community is whether LLMs genuinely reason or primarily recall memorised training data [7], [8]. High accuracy on knowledge-intensive benchmarks like MMLU could, in principle, reflect either pattern. GPQA was specifically designed to be “Google-proof,” requiring integrative reasoning over graduate-level scientific content that

cannot be answered by simple retrieval [6]. It therefore serves as a strong proxy for whether a model is truly reasoning.

B. Problem Statement

Given a vector of reasoning and knowledge benchmark scores for an LLM, can we predict whether that model achieves above-median performance on GPQA? We frame this as binary classification:

$$y = 1[\text{gpqa_acc} > \text{median}(\text{gpqa_acc})]$$

where $y = 1$ denotes a model in the top half of the GPQA distribution. Critically, GPQA accuracy is *held out* from the feature set, ensuring the label cannot be reconstructed from inputs and eliminating data leakage.

C. Related Work

Prior work has studied benchmark contamination [8] and the difficulty of disentangling memorisation from reasoning [7]. Benchmark construction efforts such as BIG-Bench Hard [2] and MuSR [4] explicitly target multi-step reasoning problems resistant to recall-based shortcuts. To our knowledge, no prior work has used cross-benchmark prediction as a structured test of reasoning-to-knowledge generalisation across a large corpus of publicly released models.

II. THEORETICAL AND CONCEPTUAL STUDY

A. Logistic Regression

Logistic regression models the posterior probability of the positive class as:

$$\hat{y} = \sigma(\mathbf{w}^\top \mathbf{x} + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}} \quad (1)$$

where $\mathbf{w} \in \mathbb{R}^d$ are learned weights and b is a scalar bias. Predictions are obtained by thresholding at 0.5.

Numerically Stable Sigmoid. For large negative inputs, the naive formulation of (1) suffers from floating-point underflow. We use the piecewise implementation:

$$\sigma(z) = \begin{cases} \frac{1}{1 + e^{-z}} & z \geq 0 \\ e^z & z < 0 \end{cases} \quad (2)$$

which avoids overflow in both branches.

B. Loss Function with L2 Regularisation

Training minimises the regularised binary cross-entropy:

$$\mathcal{L}(\mathbf{w}, b) = -\frac{1}{n} \sum_{i=1}^n [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] + \frac{\lambda}{2} \|\mathbf{w}\|_2^2 \quad (3)$$

where λ is the L2 regularisation coefficient. The L2 term penalises large weights, reducing overfitting on the moderately sized dataset. Note that the bias b is not regularised.

C. Mini-Batch Stochastic Gradient Descent

Parameters are updated via mini-batch SGD. For a batch $\mathcal{B} = \{(\mathbf{x}_i, y_i)\}_{i=1}^m$, the gradients are:

$$\nabla_{\mathbf{w}} \mathcal{L} = \frac{1}{m} \mathbf{X}_{\mathcal{B}}^{\top} (\hat{\mathbf{y}} - \mathbf{y}) + \lambda \mathbf{w} \quad (4)$$

$$\nabla_b \mathcal{L} = \frac{1}{m} \sum_{i \in \mathcal{B}} (\hat{y}_i - y_i) \quad (5)$$

Weight updates follow:

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla_{\mathbf{w}} \mathcal{L}, \quad b \leftarrow b - \eta \nabla_b \mathcal{L} \quad (6)$$

where η is the learning rate. Weights are initialised using He initialisation: $\mathbf{w} \sim \mathcal{N}(0, 2/d)$, appropriate for sigmoid activations.

D. Feature Normalisation

Features are standardised per-feature using training-set statistics:

$$\tilde{x}^{(j)} = \frac{x^{(j)} - \mu_j^{\text{train}}}{\sigma_j^{\text{train}}} \quad (7)$$

where μ_j^{train} and σ_j^{train} are computed on the training split only and applied to validation and test sets. This prevents test-set information from leaking into training via the normalisation statistics.

E. Threshold Baseline

As a reference model, we implement a rule-based threshold classifier that scans 50 evenly spaced thresholds over the range of a margin feature and selects the threshold maximising training accuracy. This requires no gradient-based learning and provides a performance floor that the logistic regression must surpass to demonstrate genuine learning.

F. Hyperparameter Selection

Grid search is performed over:

- Learning rate $\eta \in \{0.001, 0.01, 0.1\}$
- L2 coefficient $\lambda \in \{0.0, 0.01, 0.1\}$

yielding 9 configurations. The configuration maximising validation F1 is selected, and the chosen model is evaluated exactly once on the held-out test set.

TABLE I
LOGISTIC REGRESSION GRID SEARCH: VALIDATION METRICS

η	λ	Acc.	F1	AUC
0.001	0.00	0.7638	0.7541	0.8676
0.001	0.01	0.7717	0.7603	0.8708
0.001	0.10	0.8031	0.7899	0.8795
0.010	0.00	0.8740	0.8667	0.9449
0.010	0.01	0.8819	0.8760	0.9382
0.010	0.10	0.8189	0.8034	0.9023
0.100	0.00	0.9055	0.9016	0.9683
0.100	0.01	0.8740	0.8667	0.9422
0.100	0.10	0.8110	0.7966	0.9062

III. DATASET AND EXPERIMENTAL SETUP

A. Data Collection

Result files were downloaded from the Open LLM Leaderboard v2 dataset hosted on Hugging Face [1]. For each model, scores on five benchmark tasks were extracted: BIG-Bench Hard (BBH) [2], MATH-Hard [3], MuSR [4], MMLU-Pro [5], and GPQA [6]. Models missing any of these scores were discarded, yielding a final dataset of **995 models**.

B. Feature Engineering

The following 15 features are used:

- **Aggregate reasoning scores:** bbh_acc, math_acc, musr_acc, mmlu_pro_acc
 - **BBH subtask scores (10):** boolean expressions, causal judgement, date understanding, logical deduction (3/5/7 objects), navigate, object counting, penguins in a table, web of lies
 - **Score variance** across all five aggregate benchmarks
- gpqa_acc is deliberately excluded from features as it constitutes the label. Additionally, derived aggregates (reasoning_score, knowledge_score, acc_gap, total_score) are excluded to prevent indirect leakage.

C. Train/Validation/Test Split

The dataset is split with stratified sampling:

- **Test:** 15% (150 samples), held out until final evaluation
- **Validation:** 15% of remainder (127 samples), used for model selection
- **Train:** remaining 718 samples

The label is balanced: 498 negative, 497 positive (49.95% positive rate).

IV. RESULTS AND ANALYSIS

A. Grid Search Results

Table I reports validation metrics for all 9 hyperparameter configurations.

The best configuration ($\eta = 0.1$, $\lambda = 0.0$) achieves a validation F1 of 0.9016. Larger learning rates converge faster on this dataset, suggesting the loss surface is well-conditioned. Interestingly, L2 regularisation does not improve performance at the best learning rate, indicating that the model is not significantly overfitting given 718 training samples and 15 features.

TABLE II
FINAL TEST SET RESULTS

Model	Acc.	F1	Precision	Recall
Logistic Regression	0.9133	0.9091	0.9559	0.8667
Threshold Baseline	0.5000	0.0000	0.0000	0.0000

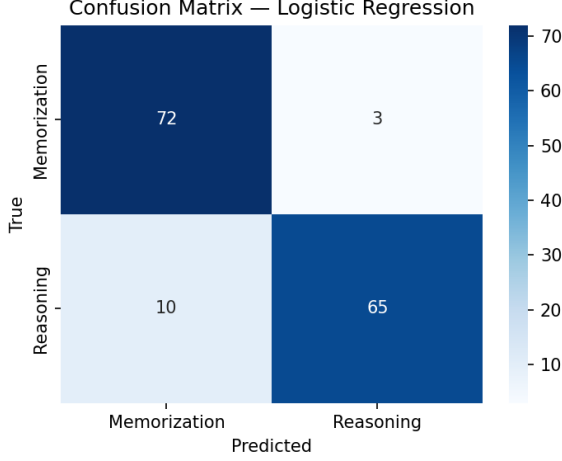


Fig. 1. Confusion matrix on the held-out test set (150 samples). Rows: true label. Columns: predicted label.

B. Final Test Set Evaluation

Table II compares the best logistic regression model against the threshold baseline on the held-out test set (150 samples).

The ROC-AUC of the logistic regression model is **0.9712** vs. 0.5000 for the baseline, confirming strong discriminative ability. The baseline degenerates to predicting the majority class because the `logit_margin` feature used by the threshold rule is absent from the leakage-free feature set, confirming that the logistic regression’s performance reflects genuine learned structure rather than a simple heuristic.

C. Per-Class Analysis

The confusion matrix (Fig. 1) and per-class metrics reveal that the model achieves higher precision (0.956) than recall (0.867) on the positive class. It is slightly more conservative in predicting “high GPQA” performance, misclassifying 10 positive instances as negative. This asymmetry is acceptable for a research setting where false positives (incorrectly labelling a mediocre model as high-performing) are more costly than false negatives.

D. Feature Weight Analysis

Table III lists the top and bottom learned feature weights. Several findings are notable:

- **MATH-Hard dominates positively** ($w = +8.45$). Mathematical reasoning is the strongest single predictor of GPQA performance. This aligns with the hypothesis that formal reasoning capabilities transfer across domains.

TABLE III
TOP AND BOTTOM FEATURE WEIGHTS (BEST MODEL)

Feature	Weight
math_acc	+8.446
bbh_acc	+2.354
musr_acc	+1.862
bbh_sub_logical_deduction_5obj	+1.269
score_variance	+1.090
bbh_sub_penguins_in_a_table	−0.533
bbh_sub_logical_deduction_7obj	−0.521
bbh_sub_web_of_lies	−0.426
bbh_sub_date_understanding	−0.411
mmlu_pro_acc	−9.504

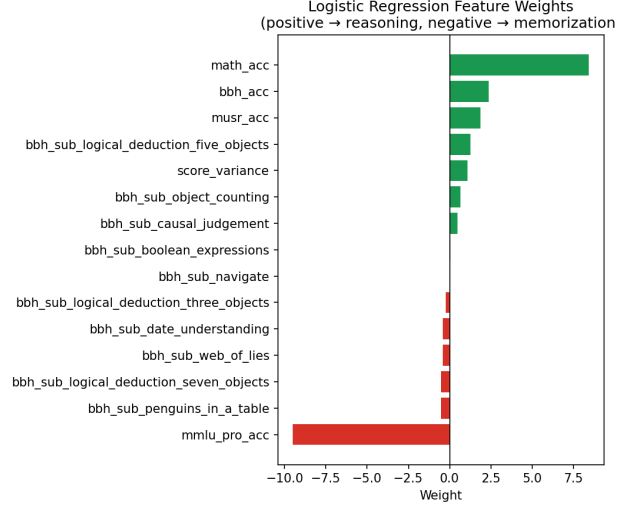


Fig. 2. Learned logistic regression weights. Green bars push toward high GPQA prediction; red bars push toward low GPQA prediction.

- **MMLU-Pro dominates negatively** ($w = -9.50$). Models with disproportionately high MMLU-Pro scores relative to their reasoning scores tend to score below median on GPQA. This suggests a genuine tradeoff: optimising for broad factual coverage does not translate to deep integrative reasoning.
- **Score variance is positive** ($w = +1.09$). Models with uneven benchmark profiles, scoring very high on some tasks and lower on others, tend to perform better on GPQA, potentially reflecting specialisation in depth over breadth.
- **BBH subtask split**: logical deduction with 5 objects is positive (+1.27), while 7-object deduction is negative (−0.52), suggesting that performance on harder subtask variants does not uniformly predict GPQA ability.

E. Training Dynamics

Fig. 3 shows the training and validation loss curves for the best model over 500 epochs. Both curves converge smoothly without divergence, and the gap between training and validation loss remains small, confirming the absence of significant overfitting.

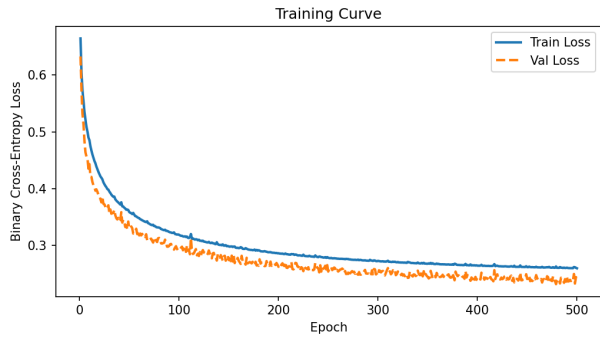


Fig. 3. Binary cross-entropy loss over 500 training epochs for the best model ($\eta = 0.1$, $\lambda = 0.0$). Validation loss tracks training loss closely.

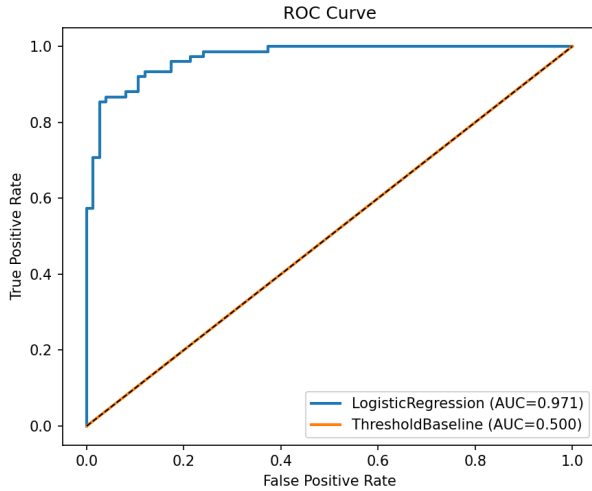


Fig. 4. ROC curves. Logistic regression AUC = 0.9712 vs. threshold baseline AUC = 0.50.

F. ROC Curve Comparison

Fig. 4 plots the ROC curves for both models. The logistic regression curve hugs the top-left corner (AUC = 0.9712), while the baseline lies on the diagonal (AUC = 0.50), confirming that the classifier captures genuine signal in the feature space.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

This work demonstrates that an LLM’s performance on graduate-level knowledge reasoning (GPQA) can be predicted with high accuracy (91.33%, AUC 0.9712) from its scores on reasoning-focused benchmarks, using a logistic regression classifier trained entirely from scratch in NumPy. The classifier significantly outperforms a rule-based threshold baseline, confirming that the learned feature weights capture meaningful structure in the benchmark performance space.

Two methodological practices were critical to validity: (1) holding out the target benchmark (GPQA) from the feature set to prevent label leakage, and (2) fitting normalisation statistics on the training split only to prevent test-set contamination

via standardisation. These precautions are straightforward but frequently overlooked in benchmark-prediction pipelines.

The most actionable finding is the strong positive weight of MATH-Hard and the strong negative weight of MMLU-Pro. Models that excel at multi-step mathematical reasoning generalise to GPQA, while models that achieve high MMLU-Pro scores without commensurate reasoning ability do not.

B. Future Work

Several directions can extend this work:

- **Richer models.** A neural network with one or two hidden layers may capture non-linear interactions between benchmark scores that logistic regression misses.
- **Temporal analysis.** The leaderboard evolves continuously. Tracking whether the reasoning-to-GPQA relationship strengthens or weakens over model generations is an interesting longitudinal question.
- **Additional benchmarks.** Including HumanEval [9] or other coding/reasoning benchmarks as features may improve predictive power and shed further light on cross-domain transfer.
- **Model-family effects.** Controlling for model architecture or training recipe (e.g., instruction-tuned vs. base models) may reveal family-level effects obscured by the current aggregate analysis.
- **Calibration.** Platt scaling or isotonic regression applied post-hoc could improve the reliability of probability estimates for downstream decision-making.

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